

# Statistical Analysis of Retail Data

A comprehensive examination of sales patterns, profitability metrics, and customer behavior in retail operations





# Executive Summary

This statistical analysis examines key retail metrics including sales, profit, discount patterns, and customer behavior. The analysis reveals critical insights about revenue concentration, profitability challenges, and pricing strategy effectiveness.



## Data Skewness

Extreme right-skewness indicates revenue dominated by few large orders



## Profitability Issue

24% of orders operating at a loss requires immediate attention



## Discount Impact

Negative correlation shows discounts eroding profit margins

# Sales Distribution Analysis

## Descriptive Statistics

\$246....

Mean Sales

Average revenue per  
transaction

\$85

Median Sales

Middle value in ordered  
data

\$13

Mode Sales

Most frequent transaction  
value

The substantial gap between mean and median sales reveals **highly right-skewed data**. This indicates that a small number of very large orders are significantly inflating the average, while most transactions are actually low-value purchases. The mode at \$13 suggests that the most common transaction size is minimal.



# Measuring Sales Variability

## Dispersion Metrics

\$487.57

Standard Deviation

Absolute measure of variation

1.98

Coefficient of Variation

Relative variability measure

The standard deviation of **\$487.57** combined with variance of 237,721.75 indicates **extremely high variation** in sales. The coefficient of variation exceeding 1.0 confirms that sales are highly inconsistent, with business performance heavily dependent on a few large orders rather than consistent transaction volume.



# Profitability Analysis

## Profit Statistics

\$28.64

Mean Profit

Average profit per  
transaction

\$174.42

Standard  
Deviation

Profit variability measure

8.13

Skewness

Distribution asymmetry

176.72

Kurtosis

Outlier concentration

Profit shows **extreme right-skewness (8.13)** and very high kurtosis (176.72), indicating that while most orders yield modest profit, a few generate exceptionally high returns. However, the large standard deviation reveals **unstable profitability** across the transaction portfolio.

# Correlation Analysis

## Relationship Between Key Metrics

1

Sales ↔ Profit: 0.49

Moderate positive relationship – higher sales generally lead to higher profit, though not consistently

2

Discount ↔ Profit: -0.32

Negative relationship – increased discount levels directly reduce profit margins

3

Quantity ↔ Sales: 0.31

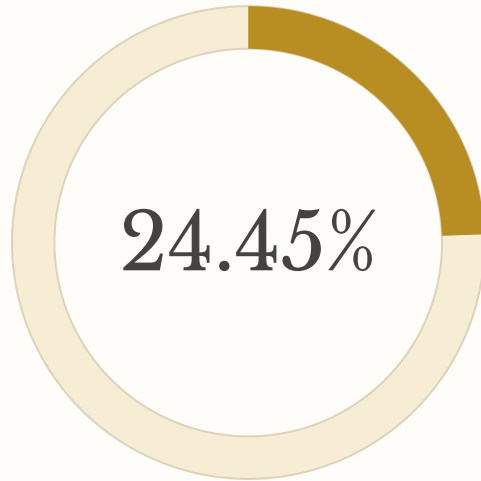
Weak to moderate relationship – larger order quantities slightly increase revenue

4

Shipping ↔ Profit: 0.36

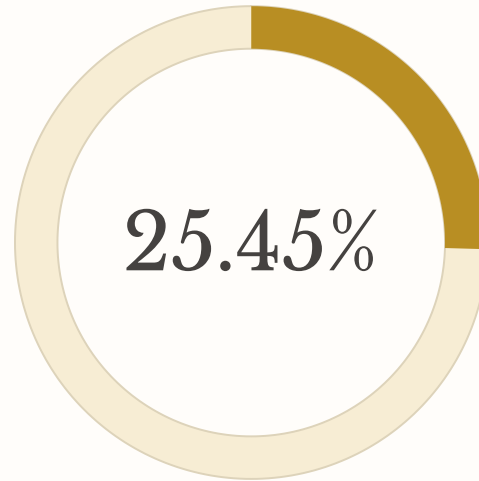
Weak positive relationship – higher shipping costs correlate with slightly improved profitability

# Probability Analysis



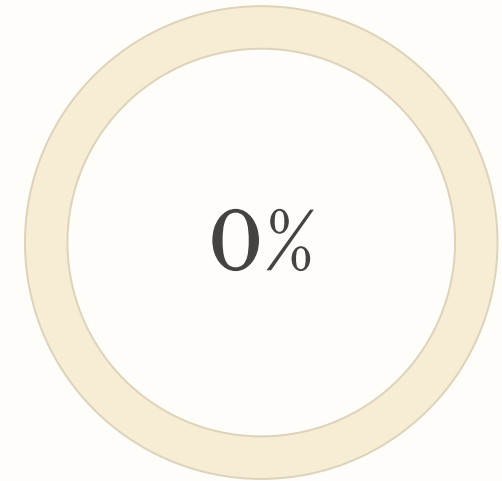
Probability of Loss

Orders with negative profit



Above-Average Sales

High-value transactions



High Discount Usage

Discounts over 20%

Perhaps most concerning, **almost 1 out of 4 orders (24.45%)** operates at a loss. This represents a serious profitability issue requiring immediate attention. Meanwhile, only 25% of orders exceed average sales value, confirming revenue concentration. Interestingly, discounts above 20% are rarely used, yet even moderate discounting significantly impacts profit margins.

# Critical Insights

## Revenue Concentration

Sales data shows extreme skewness where a few large orders dominate total revenue, creating inconsistent cash flow and revenue volatility

## Profitability Crisis

24% of all orders generate losses, indicating fundamental issues with pricing strategy, cost structure, or product mix

## Discount Strategy Backfiring

Negative correlation ( $-0.32$ ) between discount and profit reveals that current discount practices are eroding profitability rather than driving volume

## High Instability

Coefficient of variation at 1.98 indicates sales are highly unstable, making forecasting and planning extremely challenging

## Non-Normal Distribution

Extreme kurtosis (176.72) and skewness (8.13) mean data is not normally distributed, requiring use of median over mean for analysis



# Strategic Recommendations

01

## Address Loss-Making Orders

Analyze the 24% of loss-making transactions to identify root causes and implement corrective measures immediately

02

## Optimize Discount Strategy

Reduce discount frequency and magnitude given negative profit correlation, focusing on value-based pricing instead

03

## Diversify Revenue Base

Develop strategies to increase high-value transactions from current 25% to reduce dependency on large orders

04

## Improve Data Quality

Address extreme outliers and non-normal distribution through data cleaning and segmentation for better analysis



# Conclusion

Statistical analysis of retail data reveals significant challenges in profitability and revenue consistency. The data shows extreme skewness with **8.13 skewness coefficient**, indicating that revenue is dominated by a few large orders rather than consistent transaction volume. Most critically, **24% of orders operate at a loss**, representing a serious profitability crisis requiring immediate intervention.

The negative correlation of **-0.32 between discount and profit** demonstrates that current discount strategy is actively harming profitability rather than driving growth. Combined with extremely high variability (CV = 1.98), these findings suggest that business operations lack consistency and strategic pricing discipline.

**Next steps:** Conduct root cause analysis of loss-making transactions, review pricing and discount policies, and develop strategies to increase transaction value consistency.